

PREDICTING CUSTOMER CHURN AT SYRIATEL

A DATA-DRIVEN RETENTION STRATEGY

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THE BUSINESS PROBLEM

- High customer churn leading to revenue loss
 - Marketing spend replaces lost customers instead of driving growth
 - Many acquired customers also churn

WHY CHURN MATTERS

- Customer retention is cheaper than acquisition
 - Missed churners represent permanent loss
 - Blanket retention strategies waste resources



PROJECT OBJECTIVE

- Predict customers most likely to churn
 - Identify key churn drivers
 - Compare models based on business impact
 - Recommend targeted retention actions

DATA OVERVIEW

- 3,333 customer records
 - Usage behavior, service plans, support interactions
 - Binary churn target (Yes / No)
 - Usage behavior, service plans, support interactions
 - No missing values

MODELING STRATEGY

- Train-test split with stratification
 - Compare multiple classification models
 - Focus on business cost of errors

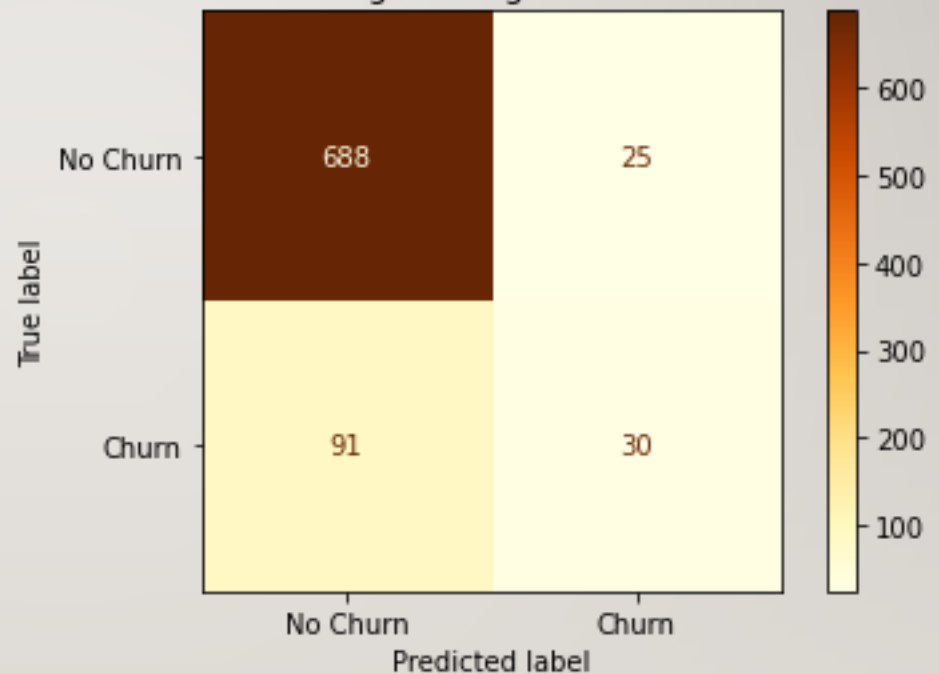
BASELINE MODEL

- Dummy model predicts all customers as 'No Churn'
 - Churn recall: 0%
 - Shows need for predictive modeling

LOGISTIC REGRESSION

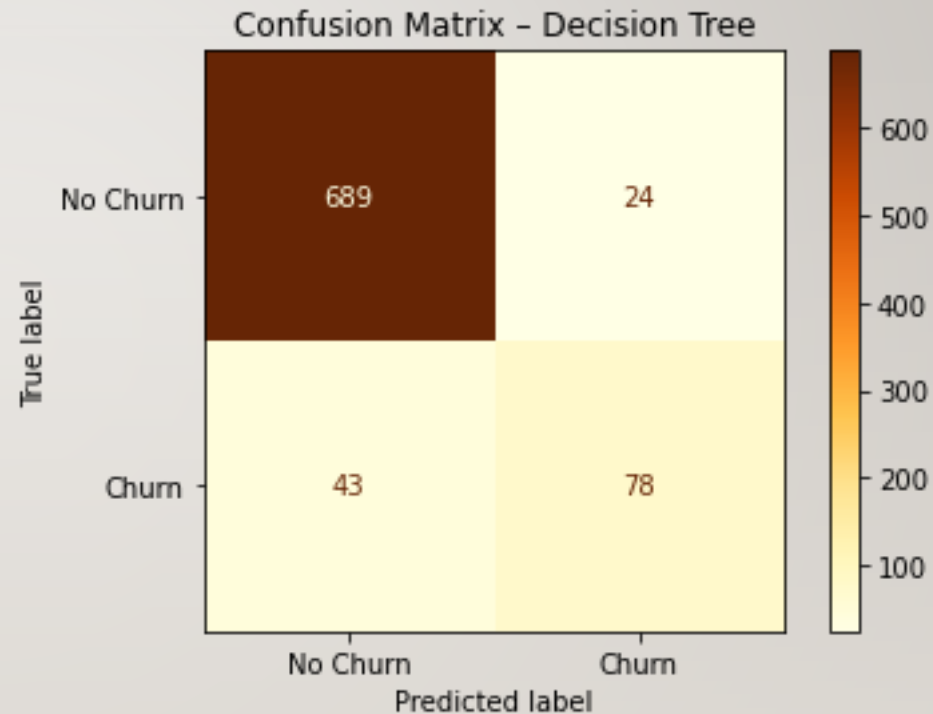
- Default threshold misses many churners
 - Threshold tuning improves recall
 - Still limited in capturing churn patterns

Confusion Matrix - Logistic Regression (Default Threshold)



DECISION TREE MODEL

- Correctly identifies ~64% of churners
 - Fewer false negatives
 - Clear, interpretable decision rules



MODEL COMPARISON

- Baseline: No churners identified
 - Logistic Regression: Improved but inconsistent
 - Decision Tree: Best balance of recall and precision

KEY CHURN DRIVERS

- Frequent customer service calls
 - International plan subscriptions
 - Short account tenure and usage patterns

BUSINESS RECOMMENDATIONS

- Focus on retaining high-risk customers instead of broad customer acquisition.
- Prioritize frequent customer service callers for faster resolution and proactive outreach.
- Review international plan pricing and billing clarity to reduce cost-driven churn.
- Introduce early engagement programs for new customers with short account tenure.
- Integrate churn predictions into marketing and customer service workflows.

CONCLUSION

- SyriaTel's churn problem can be effectively addressed using predictive analytics.
- The decision tree model accurately identifies customers at risk of churn.
- Targeted retention strategies are more cost-effective than continuous customer replacement.
- Ongoing model monitoring will ensure sustained impact over time.